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Review

Agricultural water management in water-starved countries: challenges and opportunities

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Abstract

Agriculture commands more water than any other activity on this planet. Although the total amount of water made available by the hydrologic cycle is enough to provide the world's current population with adequate freshwater, most of this water is concentrated in specific regions, leaving other areas water-deficient. Because of the uneven distribution of water resources and population densities worldwide, water demands already exceed supplies in nearly 80 countries with more than 40% population of the world. Consequent to future population increase in these countries, supplies of good-quality irrigation water will further decrease due to increased municipal–industrial–agricultural competition. These facts reveal that the time has come for the sustainable management of available water resources based on global, regional, and site-specific strategic options: (1) understanding the concept of 'virtual water' and potential use of this water as a global solution to regional deficits, i.e. the water-short countries may import a portion of food crops or other commodities that require more water and export those that need less water in production; (2) improvement in current efficiencies of agricultural water use and conservation, both in the rain-fed and irrigated agriculture, i.e. to produce more with the existing resources with minimum deterioration of land and water resources; (3) use of efficient, economic, and environmentally acceptable methods for the amelioration of polluted waters and degraded soils, and (4) re-use of saline and/or sodic drainage waters via cyclic, blended, or sequential strategies for crop production systems, wherever possible and practical. We believe that these strategies will serve as the four pillars of integrated agricultural water management and their suitable combinations will be the key to future agricultural and economic growth and social wealth, particularly

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in regions that are deficient in freshwater supplies and are expected to become more deficient in future.
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1. Introduction

Water, it is said, may become as precious as oil during this century. Even though the total amount of water made available by the hydrologic cycle is enough to provide the world's current population with adequate freshwater, most of this water is concentrated in specific regions leaving other areas water-deficient (Pimentel et al., 1999). Because of the uneven distribution of water resources and population densities worldwide, water demands already exceed supplies in nearly 80 countries with more than 40% population of the world (Bennett, 2000). The minimum average annual amount of water required per capita for food production is 0.4×10^6 l (Postel, 1996), which is about four-times less than is consumed in the United States (1.7×10^6 l). Similarly, the minimum daily per capita water requirement of 50 l for human health, including drinking water, is eight-times less than is used in the United States (Gleick, 1996). Between 1960 and 1997, per capita availability of freshwater worldwide declined by about 60%. Another 50% decrease in per capita water supply is projected by the year 2025 (Hinrichsen, 1998).

The amount of liquid freshwater compared with the world's total water is just like a spoon of water in about 1 l of water. This is because of the fact that almost 97% of the world's water occurs in the oceans (Turner, 2001), which has an electrical conductivity (EC) around 55 dS m^{-1} (total dissolved solids $\approx$ 35 000 mg l^{-1}) and sodium (Na^+) concentration more than 450 mmol l^{-1} (Suarez and Lebron, 1993). Of the liquid freshwater, more than 98% occurs as groundwater whereas less than 2% occurs in more visible form of streams and lakes, which often are fed by the groundwater. Excess rainfall, which may be defined as total precipitation minus surface runoff and evapotranspiration, infiltrates deeper into the soil and eventually percolates down to the groundwater formations or aquifers (Bouwer, 2000a). The amount of precipitation that contributes to groundwater is principally impacted by the climatic conditions. For instance, about 30–50% of the precipitation contributes to groundwater in temperate and humid climates. It ranges from 10 to 20% in the Mediterranean climate. The amount of precipitation ending up in groundwater is the lowest in hot and dry climates, which may be as little as 2% or even less (Tyler et al., 1996; Bouwer, 2002a). Although accurate natural recharge rates are difficult to predict (Stone et al., 2001), the above estimates reveal the approximate amounts of water that can be pumped from the aquifers under different climates without depleting the groundwater resources. As a general practice in several arid and semiarid regions of the world where groundwater is the main water resource, pumping greatly exceeds recharge resulting in lowering the groundwater levels at alarming rates (Pimentel et al., 1999). Of particular concern is the decline in the water table in two of Asia's major breadbaskets, namely, the Punjab of Indian subcontinent and the North China Plains (Seckler et al., 1999). The situation is even worse in the Middle

East and parts of Africa, which ran out of water decades ago in the sense that they were unable to meet their food requirements from the available water resources. Several other countries have recently entered water deficit (Allan, 2001).

Agriculture is the largest single user of water with 65–75% of freshwater being currently used for irrigation (Bennett, 2000; Prathapar, 2000). In some cases, it draws as much as 90% of the total water (Allan, 1997). The following factors, either alone or in different combinations, have contributed and may continue to affect the availability of good-quality irrigation water in different regions of the world. (1) Inherited shortage of water in certain areas as a result of their geographical location where rainfall is very low, groundwater use is not feasible due to economic, political and/or technical reasons, water treatment options have economic limitations, and transportation of good-quality water from other areas is not practical. (2) Increased cropping intensities on already cultivated lands consuming more water per unit area cultivated, i.e. vertical expansion of irrigated agriculture, which has simultaneously resulted in degradation of the land and associated water resources at some places. (3) Cultivation of crops on new lands requiring additional amount of water, i.e. horizontal expansion of irrigated agriculture. Such expansion has deteriorated surface and groundwater quality at places where marginal lands were brought under cultivation without appropriate management practices. (4) Increased industrial and domestic use of good-quality water as a result of an increase in population coupled with higher living standards. The present world population of about six billion is generally projected to increase in the range of 25–80% during the next 50 years. Most of the projected global population increases are expected to take place in the Third World countries that already suffer from water, food, and health problems. (5) Contamination of surface and groundwater resources by a variety of point and non-point pollution sources.

Since freshwater has always been an integral component of food production, it is obvious that the water requirements associated with producing food for the future world population are huge. It is, therefore, apparent that strategic water management will be the key to future agricultural and economic growth and social wealth, both in developed and developing countries. This paper explores the possible options regarding the sustainable agricultural water management to fulfil the future food requirements in areas that are already deficient in freshwater supplies and are expected to become more deficient in future.

2. Potential use of virtual water

As developed by Professor J.A. Allan, the concept of ‘virtual water’ has been introduced for water-short countries in recent years. These countries can minimize their use of water and achieve food security at the same time by importing a portion of their food requirements from other areas or countries where water resources are adequate and available at a lower cost (Allan, 1996). Trading of other water-intensive commodities such as hydro-electric power falls in the domain of this concept in a similar way. The nations receiving food and electric power not only get the commodities but also the water that is necessary to produce them. Since this water is ‘virtually’ embedded in the commodity, it is called ‘virtual water’ (Allan, 1999a).

Table 1
Estimates of water needed to produce different food items in the Middle East and North Africa region

Food item	Amount of water needed[a] ($m^3\,Mg^{-1}$)
Pulses	1000
Citrus fruits	1000
Roots and tubers	1000
Cereals[b]	1500
Palm oil	2000
Meat poultry fresh	6000
Meat sheep fresh	10000
Meat bovine fresh	20000

Modified from Allan (1999a); developed from agricultural production and trade statistics of the Food and Agricultural Organization of the United Nations, Rome.

[a] Quantifying the exact volumes of water needed to produce water-intensive food items is difficult. These estimates give an idea about the amount of water needed to produce some food items.

[b] In some areas, some cereals such as wheat may need 1000 m^3 of water to produce 1 Mg of grain (Allan, 2001). On the other hand, rice usually consumes more water than other cereals and may need more than 7000 m^3 of water to produce 1 Mg of grain (Rosegrant et al., 2002).

The concept of virtual water compares the amount of water embodied in a crop that can be purchased internationally with the amount of water which would be required to produce that crop domestically. For instance, for every kilogram of wheat (*Triticum aestivum* L.) imported, the water-stressed country will also get about 1 m^3 (1000 l) of virtual water at much less cost than the price or value of the same quantity of water from the local water resources, if available, in the country itself (Allan, 1999b). This means that import of each megagram (1 Mg = 1 tonne) of wheat grain has about 1000 m^3 (1000 Mg) of virtual water embedded in it. Similar calculations can be made for other food items requiring greater amounts of water in the production process (Table 1). It is, therefore, easier and less ecologically destructive to import grain rather than to pipe 1000 times greater amount of water to produce the same commodity (Turton, 1999). The nations with scarce water resources gain by importing water-intensive commodities, while exporting goods that require less water in production. This strategy is particularly important in years when the world prices of food commodities are lower than the cost of production in water-stressed areas (Wichelns, 2001).

The role of virtual water is important in certain regions such as the Middle East and parts of Africa. The Middle East is the first region in the world that ran out of water. The water demands of the populations of the Arabian Peninsula and desert Libya had exceeded the capacity of their water resources for food sufficiency by the 1950s. Israel and Palestine also ran out of water by the same time, Jordan in the 1960s, and Egypt in the 1970s. In the past, some of these countries have attempted to become self-sufficient in food while using their own water resources. For example, Saudi Arabia—one of the major countries in both cereal production and consumption in the Middle East region—had begun to produce sufficient wheat for most of its needs in the mid 1980s and it was about to become a significant wheat exporter in the world markets (Allan, 1997). Until the early 1990s, the country used significant amounts of its fossil water to grow maize (*Zea mays* L.) (Allan, 1999a). Since fossil water is extremely pure but non-renewable, the country had to reduce crop production because it was an uneconomic way to use its fossil water. There are some other extreme

examples where water-starved countries extracted their limited water resources to take a pride of self-sufficiency, which was not sustainable (Otchet, 2000).

The hydrological system of the water-starved countries is definitely less and less able to meet the rising demands being placed on it. However, a choice exists between two development options: more today (less tomorrow) or more preservation (more tomorrow). Do the water-starved countries want large-scale extraction of groundwater resources for maximum benefit of the present generation, to take a short-term national pride of self-sufficiency, or a restricted extraction that ensures sustainable development and conservation of the resource base? With a few exceptions, most countries have chosen to go for the long-term conservation of their limited water resources, although the approaches vary to a considerable extent. This is evident from the fact that many countries in the Middle East and parts of Africa have already been implementing the virtual water strategy implicitly for several years. Since the oil boom of the 1970s, the average annual rate of growth in food imports has substantially increased in several Middle East countries. Allan (1999a) states that since the end of the 1980s, the Middle East and North Africa region has been importing 40×10^6 Mg of cereals and flour annually. He reveals that more virtual water flows into the region each year than flows down the Nile into Egypt for agriculture. Estimates show that the Middle East imported about 25% of its water requirement as virtual water at the millennium. Virtual water will provide 50% of its water requirement in 2050 (Allan, personal communication, 2002).

The trading of virtual water combines agronomic, economic, and political aspects. The agronomic component involves the amount of water used to produce crops. The virtual water perspective is consistent with the concept of integrated water management, in which many aspects of water supply and demand are considered while determining the optimal use of limited water resources (Bouwer, 2002b). There has been a need to modify cropping plans to get maximum benefits from the available but limited water resources in water-deficit countries and regions in accordance with the concept of virtual water. Apart from economic, social and political aspects, such cropping plans will depend on a number of considerations: (1) water factors such as the quantity and quality of water available to produce crops together with the preservation of fossil water at the same time; (2) soil factors including soil reaction (pH), texture, structure, salinity and sodicity levels, and nutrient availability status; (3) crop traits such as water use efficiency, adaptability to local conditions, and local and/or international market demand; (4) farming aspects such as size and productivity of agricultural farms; (5) availability of manpower, farm inputs, and farm machinery, and (6) climatic conditions, particularly air temperature and velocity, rainfall, and relative humidity. There is a research need for such aspects. In general, efforts should be made in water-stressed countries that they use good-quality water on good soils to produce high-value crops that have low water requirements. The marginal-quality waters can be used on poor soils to grow relatively low-value crops. This aspect is discussed in a later section of this paper that deals with the use of drainage waters for crop production systems.

The economic aspect of virtual water involves the opportunity cost of water, which is its value in other uses that may include production of alternative crops or use in municipal, industrial, or recreational activities. In particular, the opportunity cost of water use, which is a key component of the virtual water perspective, must be considered when seeking an efficient allocation of scarce water resources. Indeed, the economic aspect of the virtual water concept is closely related to the comparative advantage concept from international

trade theory. This concept suggests that nations should export products in which they possess a relative or comparative advantage in production, while they should import products in which they possess a comparative disadvantage. Wichelns (2001) has exemplified the crop production and international trade data for Egypt where imports of wheat and maize provide substantial amounts of virtual water and land that would otherwise be required to replace those imports. Wheat imports have grown from 1.2×10^6 Mg in 1961 to 7.4×10^6 Mg in 1998, making the country the third largest importer of wheat after China and Russia. Maize imports have grown from 0.1×10^6 Mg in 1961 to 3.1×10^6 Mg in 1998. However, the exports of rice (*Oryza sativa* L.) and the virtual water embedded in those exports have been increasing in recent years as a result of expansion in rice production in response to agricultural policy reforms. Exports of Egyptian cotton (*Gossypium barbadense* L.), which have historically been an important source of foreign exchange in Egypt, have declined over time. The decline in cotton exports is attributed to a decreased production as a result of changes in agricultural policies and government decisions regarding allocation of the crop between domestic and international markets. At this stage, the policies that encourage farmers to consider the opportunity cost of water used in rice production would be helpful in motivating them to use water more efficiently and to choose alternative crops that need less water in production. Greater production and processing of Egyptian cotton, certain fruits and vegetables for export would improve rural incomes and enhance food security. However, time is required to change people's perception regarding the potential use of virtual water. The reasons are that irrigation water is almost free in Egypt, about 40% of the labor force works in agriculture, and most farmers have small holdings of <2 ha.

Contrary to Egypt, Israel—a severely water-deficient country that ran out of water nearly half a century ago—has been able to implement a more sustainable water policy. Its farmers have the means to employ the most efficient irrigation systems. Israel is one of the few countries in the world to charge a high proportion of the delivery cost (40%) for irrigation water (Allan, 1999a). Despite needing up to four times more water than is available, Israel has been able to adopt the virtual water development strategy to balance its water budget through easy access to water that is embedded in cereals imported from water-rich countries (Jobson, 1999). Some other countries of the region, such as Jordan, Tunisia, and Morocco have also started to take the same approach (Allan, 2001).

An important economic aspect of virtual water trade relates to the fact that the water embedded in grain is sometimes traded at less than its production cost. For instance, the grain price increased in 1996 to US$ 240 Mg^{-1} but fell back to US$ 140 Mg^{-1} by May 1997 in the international market. The cost of producing wheat was estimated to be about US$ 200 Mg^{-1} (Allan, 1998). In addition, there has been a decreasing trend in world prices of several food commodities during the last three decades. Between 1970 and 2000, the international prices of three major cereals—wheat, rice and maize—fell in the range of 50–60%. During the next two decades, the world prices of most cereals are projected to decline, but more slowly when compared to the previous trend. For instance, the international price of rice is projected to decrease from US$ 285 Mg^{-1} in 1995 to US$ 221 Mg^{-1} in 2025. The trend for wheat will be from US$ 133 Mg^{-1} to US$ 119 Mg^{-1}. The price of maize will be little affected as it is projected to be US$ 104 Mg^{-1} in 2025 compared to US$ 103 Mg^{-1} in 1995. The prices of other coarse grains will decrease from US$ 97 Mg^{-1} to US$ 82 Mg^{-1} during the next two decades (Rosegrant et al., 2002). This means that economies of the water-stressed countries

that import grains may get a subsidized bargain. Therefore, water-deficit economies may receive a double benefit through accessing embedded virtual water at an advantageous price.

For political leaders of the water-starved countries, political imperatives are more important and, therefore, are more compelling than scientific facts. The same applies to the water issues in their countries; the political imperatives lead them to assert that their economies have not run out of water. On this situation, Allan (1999a) comments "It would require an inhuman level of courage for a political leader of a country that has enjoyed water security for 5000 years to announce that supplies are no longer adequate. Instead, leaders insist that supplies are 'sufficient'. But this is deceptive. Supplies are 'sufficient' for the small amounts needed for drinking: one cubic meter per year per person. They may also cover current domestic and industrial needs, although both are on the rise. But there isn't enough freshwater to cover these demands in addition to the tremendous amounts needed for food production." Therefore, instead of paying the political costs of publicly recognizing this fact, leaders rely on the convenient solution of virtual water. As a result, trading of virtual water embedded in food and other commodities seems to be a very good political step to achieve peaceful solutions to water conflicts within water-deficient countries, and between water-deficient and water-sufficient regions and countries. In addition, it could be used effectively to avoid dealing with a very real problem.

Many water-short countries will continue to rely on imported food crops to provide a significant portion of their food supply, while also producing a portion of their food requirements domestically. Thus, the import of virtual water via imported food and export of other commodities requiring less water will remain a valid concern for the water-short nations seeking to maximize the value of their limited water resources. The land, labor, and capital embodied in agricultural imports and exports must also be considered in countries where one or more of those resources are limited, or where reducing unemployment is an important policy goal (Wichelns, 2001). In such countries, labor-intensive crop production and processing strategies will be desirable. In addition, an important aspect of virtual water trading is that it should not be used as a political weapon, rather it should be internationally controlled and treated in a concept similar to the movement of petroleum products from oil-rich to oil-poor countries. Bouwer (2000a) has suggested that in addition to the Organization of Petroleum Exporting Countries (OPEC), we may then have an Organization of Food Exporting Countries (OFEC) with international controls and representation of the food importing countries. However, this will need a global understanding among both the virtual water importing and exporting countries.

3. Efficient water use and conservation strategies

Despite limitations with the supply of freshwater in several regions, considerable amounts of water are lost through one or any combination of the mechanisms such as: (1) evaporation from soil surface during conveyance and irrigation, (2) leakage during storage and transport to the fields where crops are grown, (3) runoff, and (4) uncontrolled drainage. Under irrigated agriculture, about 30% of water to be used as irrigation is lost in storage and conveyance. There are also other losses such as runoff and drainage when this remaining 70% water reaches the farmers' fields. Postel (1993) has estimated the worldwide irrigation

Table 2
Approximate values of the water-use efficiencies of irrigated and rain-fed agriculture in semi-arid areas

Potential water losses and uses	Irrigated agriculture (fraction of available water) (%)[a]	Rain-fed agriculture[b] (fraction of rainfall) (%)
Storage and conveyance	30	0
Runoff and drainage	44	40–50
Evaporation (from soil or water)	8–13	30–35
Total water losses[c]	82–87[d]	70–85[d]
Water used as transpiration	13–18	15–30[e]

Modified from Wallace (2000).

[a] Rainfall and stored surface or groundwater.

[b] Based on the data given in Wallace and Batchelor (1997).

[c] Some of the water "lost" from an irrigated field may return to aquifers or streams from which it can be extracted again, provided the necessary infrastructure is available and the water quality has not deteriorated beyond acceptable limits.

[d] Calculated as sum of water lost through (1) storage and conveyance, (2) runoff and drainage, and (3) evaporation (from soil or water).

[e] Under typical conditions of farmers' fields in sub-Saharan Africa, the amount of rainfall used as transpiration may be much lower with a range of 4–9% (Rockstrom, 1999).

efficiency, i.e. the amount of water used as evapotranspiration compared to the amount of water delivered to the field, to be about 37%. This estimate suggests that about 63% of the water delivered to the field is lost as runoff, drainage, or both. This means that in addition to 30% of water wasted in storage and conveyance, about 44% of the total water available at the source is lost as runoff and/or drainage. Wallace (2000) suggests that some of the water "lost" from an irrigated field may return to aquifers or streams from which it can be extracted again, provided the necessary infrastructure is available and the water quality has not deteriorated beyond acceptable limits. A summary of the current efficiency with which water is used in both rain-fed and irrigated agriculture is given in Table 2. This estimate reveals that globally only 13–18% of the initial water resource is used as transpiration by a crop in irrigated agriculture. Under rain-fed conditions of West Africa (infrequent, but intensive rainfall with the tendency of sandy soils to form crusts resulting in low infiltration rates), Wallace and Batchelor (1997) estimated that transpiration could use 15–30% of the rainfall in case of research trials. Under typical farmers' fields in the region, Rockstrom (1999) estimated transpiration to be 4–9% of the rainfall. The amount of water transpired is important because it reflects the amount of water that passes through a crop and is associated with the crop growth and yield as a measure of water-use efficiency.

Although low irrigation and water-use efficiencies may seem disappointing, the fact that they are so low provides a scope for improvement. For instance, the amount of rainfall used as transpiration in some parts of sub-Saharan Africa is estimated to be 5%. If this amount could be increased from 5 to 10%, then the vegetation yield in this region could be doubled. This is not an unreasonable target (Wallace, 2000). Oba et al. (2000) have compared different types of vegetation growth in arid regions of sub-Saharan Africa, which depend on (1) soil moisture, structure, and water storage capacity, and (2) rainfall amount, duration, and distribution patterns over several years. As a derivative of rainfall and biomass production, rainfall use efficiency of herbaceous vegetation varied from year to year, increasing when

rainfall was increased and biomass production was greater. By contrast, the rainfall use efficiency of both grazed and ungrazed shrubs, such as *Indigofera spinosa* (Forsk.) Mathew, was greater during dry years than wet years. This finding suggests that the dwarf shrubs have evolved greater abilities than herbaceous vegetation to conserve water and increase photosynthetic activity under environmental stress. The shrub species is palatable by sheep, goats, camels, and donkeys of the northeastern Africa. Therefore, the plant species with high water-use efficiency and economic value may be more suitable for use under drought conditions. Similarly, it is possible to increase water-use efficiency by 25–40% through modifying practices that involve tillage and from 15 to 25% through nutrient management in soils (Hatfield et al., 2001). Recent publications (Oweis et al., 2000; Wallace, 2000; Hatfield et al., 2001; Turner, 2001) provide valuable insights into different approaches and practices that can help increase water use and water-use efficiency in both rain-fed and irrigated agriculture. Because of water scarcity and limited availability of new good-quality arable land, future increases in agricultural production will have to rely heavily on existing land and water resources. Thus, there is a great potential for improving water-use efficiency in agriculture, particularly in those areas where need is the greatest.

Water needs for irrigation can be met, in part, by practicing uniformity of water application—precise irrigation with microirrigation—that delivers water from piped mainlines and laterals directly to the root zone frequently and in small amounts, and at rates matched to crop needs. This irrigation strategy has shown to be the best method for saline waters (Shalhevet, 1994). However, such precise irrigation systems are expensive, but benefits include reduction of hidden costs of water wastage and land degradation, and the environmental costs of drainage and land reclamation (Hillel, 2000). The net benefits of microirrigation improve markedly when such advantages are taken into account. For instance, an economic analysis of irrigation systems for cotton (*Gossypium hirsutum* L.) production in California indicated that gravity-flow systems were more profitable than the pressurized systems where there was no cost or restrictions to the farmer on drainage water disposal (Letey et al., 1990). If the costs of drainage water disposal were imposed on the farmers, a point would be reached where a switch from a gravity flow to a pressurized irrigation system would be economically justified to the farmers. A tax on groundwater withdrawals in a region where demand exceeds the natural rate of recharge will have a similar impact on the relative cost of microirrigation. Thus, there is a need of widespread adoption of policies that motivate farmers to reduce off-farm impacts and encourage entrepreneurs to develop low-cost microirrigation systems that are financially compatible with a wide range of crops and production environments (Postel et al., 2001).

The present cost of water to the farmer in several irrigated regions is usually too low to have a real impact on demand, much less to actually bring supply and demand into balance. This is one of the reasons for mismanagement of the available water resources, i.e. it decreases the incentive for the farmers to use water efficiently. Shortage of water and inadequate funding for maintaining irrigation works have focused attention on the potential for water charges to generate financial resources and reduce demand for water through volume-based charges (Perry, 2001). However, in many developing countries, the facilities required for measured and controlled delivery, which are essential for volume-related charges, are not available. The introduction of such facilities would require a massive investment in physical, legal, and administrative infrastructure. Even considerable time may be required to transform the

existing form of supplies to volume-based supplies. An alternative approach to cope with shortage would focus on assigning volumes to specific uses—water rationing—has promise for several reasons including simplicity, transparency and potential to tailor allocations specifically to ambient hydrological situations.

Historically, agricultural drainage systems have been designed and managed only for crop production objectives. However, in recent years, dual goals of environmental protection and crop production have been focused. The more optimized the drainage system, the better the opportunities for managing agricultural fields for both economically optimum yields and minimum off-site environmental effects. There are several drainage-management options that can satisfy production objectives and help protect the environment. For instance, controlled drainage may be used in some areas to conserve water and reduce deficit soil water stresses (Wesström et al., 2001). In addition to conserving drainage water, controlled drainage has the benefit of reducing losses of nutrients, such as nitrate (NO_3^-), from the fields having good subsurface drainage (Skaggs and van Schilfgaarde, 1999). This approach, originally used for water quality purposes (Meek et al., 1970), has been extended to several countries for investigation. The concept is to regulate the water table in order to maintain the water level at a depth favorable for optimum crop growth. This can be achieved at certain places through design and management of drainage systems in a way that only the minimum amount of drainage water needed to satisfy the drainage needs for crop production is removed from the field (Skaggs and van Schilfgaarde, 1999). If this can be done, not only will water that can be used by the crop be conserved, but salt and nutrient loads in the drainage water will also be reduced, i.e. both the goals of crop production and water quality protection will be achieved.

There is a likelihood of more weather extremes such as more periods with excess rainfall and more periods with low rainfall that could eventually cause drought in some regions. Traditionally, dams and surface reservoirs have been constructed to store surplus water for use in times of water shortage. However, good dam sites are getting scarce and dams have several disadvantages such as interfering with the stream ecology, displacement of people from the dam sites, loss of scenic aspects and recreational uses of the rivers, increased waterborne diseases and other public health problems, evaporation losses (particularly in case of long-term storage), high cost, potential for structural problems and failure, and no sustainability because all dams lose their storage capacity when they gradually fill up with silt and other sediments (Postel, 1999; Manouchehri and Mahmoodian, 2002). Alternatively, water can be stored via artificial recharge of groundwater. Such storage may be targeted on a long-term basis. Considering that about 98% of the world's liquid freshwater supplies already exist as underground, there seems enough space to accommodate additional amount of water. Bouwer (2000a) has described different possibilities of achieving artificial recharge of groundwater with different infiltration systems and from soils of different textures, contamination levels, and stratified layers. While evaluating the potential of groundwater storage, he states: "The big advantage of the underground storage is that there are no evaporation losses from the groundwater. Evaporation losses from the basins themselves in continuously operated systems may range from 0.5 m per year for temperate humid climates to 2.5 m per year for hot dry climates. Groundwater recharge systems are sustainable, economical, and do not have the eco-environmental problems that dams have. In addition, algae which can give water quality problems in water stored in open reservoirs

do not grow in groundwater. Because the underground formations act like natural filters, recharge systems also can be used to clean water of impaired quality." A possible disadvantage with underground storage is deterioration in quality of the water to be stored in case groundwater is of poor quality. Owing to the disadvantages of dams listed above, new dams are increasingly difficult to construct except in some developing countries where the advantages of abundant and cheap hydro-electric power are more attractive than the disadvantages. Thus, for the foreseeable future, dams are expected to play an important role in the development of water resources in such countries (Schultz, 2002). However, in developed countries such as the United States, several dams have already been breached and more are scheduled for the same fate, mostly for ecological and environmental reasons.

4. Amelioration of polluted waters and degraded soils

In addition to depletion of the freshwater resources, pollution of surface and groundwater resources threatens human and animal life and other biota. Pollution of either surface or groundwater may impact the quality of each other and subsequently the soils irrigated with such waters (Bouwer, 2000b). For instance, the movement of contaminated surface water down to the deeper soil layers may impact groundwater quality. The polluted groundwater, in turn, can cause pollution of surface water when this contaminated water moves into streams where it maintains the base flow, and also into lakes and coastal waters. Salinity and sodicity are the principal water and soil quality concerns in several irrigated regions. At some places, waters may contain a variety of pesticides, and excessive concentrations of selenium (Se), boron (B), arsenic (As), and NO_3^- and a number of other trace metals (Suarez and Lebron, 1993; Ayars and Tanji, 1999). Where municipal sewage effluent is used for irrigation, particularly in untreated form, a whole new spectrum of pollutants can be added to the soil and groundwater (Bouwer, 2000b), and subsequently to the human and animal food chains (Qadir et al., 2000).

A number of point sources contribute to pollution of surface and groundwaters, which include sewage and industrial wastewater discharges, leaking ponds or tanks, and waste disposal areas, among others (Pimentel et al., 1999). On the other hand, agriculture is the main non-point polluter of groundwater. Applications of fertilizers, pesticides, and salty water may contaminate the drainage water that moves from the root zone to the underlying groundwater. Such water quality problems may be expected to increase in future as the use of agricultural chemicals will increase when efforts will further intensify to increase agricultural production. Thus, there is a need for the maintenance of water and soil quality, which is not a one-time event but rather a continuing process.

Point-source pollution is, at least in principle, relatively simple and convenient to prevent and control. A much greater threat to world's liquid freshwater resources is non-point-source pollution of groundwater (Bouwer et al., 1999). The chemical treatment of polluted waters and degraded soils for agriculture has become cost-intensive (Gleick, 1993; Pimentel et al., 1997; Qadir et al., 2001a). For instance, the treatment of highly saline ocean water is not an economic source of good-quality water needed by agriculture. The reason is that the amount of desalinized water required to grow maize on 1 ha would cost US$ 14 000, while all other inputs including fertilizers cost less than US$ 500 (Pimentel et al., 1997). This

Table 3
Metal ion hyperaccumulating plant species

Element	Plant species above-ground plant parts ($\mu g\ g^{-1}$ dry matter)	Concentration in ground biomass ($Mg\ ha^{-1}$)	Annual above-ground biomass ($Mg\ ha^{-1}$)
Cadmium	*Thlaspi caerulescens*	3000 (1)[a]	4
Cobalt	*Haumaniastrum robertii*	10200 (1)	4
Copper	*Haumaniastrum katangense*	8356 (1)	5
Lead	*Thlaspi rotundifolium* subsp.	8200 (5)	4
Manganese	*Macadamia neurophylla*	55000 (400)	30
Nickel	*Alyssum bertolonii*	13400 (2)	9
Nickel	*Berkheya coddii*	17000 (2)	18
Selenium	*Astragalus pattersoni*	6000 (1)	5
Thallium	*Iberis intermedia*	3070 (1)	8
Uranium	*Atriplex confertifolia*	100 (0.5)	10
Zinc	*Thlaspi calaminare*	10000 (100)	4

Modified from Brooks et al. (1998).

[a] Values in parentheses are equivalents for non-accumulator plants.

water treatment figure does not even include the additional cost of moving the large amount of water from the ocean to agricultural fields. Thus, it is not a matter of using the developed technologies for the treatment of such highly saline waters and their re-use for agriculture, there are economic and biophysical limitations to their use and implementation (Pimentel et al., 1999). Bouwer (2000a) has provided guidelines regarding the local re-use of municipal wastewater after a series of treatment processes. This treated effluent can be used for urban irrigation of parks, playgrounds, sports fields, golf courses, and road plantings. Other uses may include urban lakes, fire fighting, toilet flushing, and industries.

In recent years, phytoremediation—a plant-based amelioration strategy—has emerged as a low-cost and environmentally acceptable technique. It has been shown that some plants have the ability to remove significant amounts of undesirable constituents such as heavy metals from the metal-contaminated environments (Salt et al., 1998; McGrath et al., 2002). Rhoades (1999) has suggested mustard (*Brassica juncea* L.) as an effective species capable of accumulating substantial amounts of Se in its shoots. Several studies have shown *Thlaspi caerulescens* J. Presl as an efficient accumulator of cadmium (Cd) in its above-ground parts (Brooks et al., 1998; Nedelkoska and Doran, 2000; Whiting et al., 2000). Robinson et al. (1997) have found *Berkheya coddii* as an excellent nickel (Ni) hyperaccumulator with the ability to remediate moderately contaminated soils (100 $\mu g\ Ni\ g^{-1}$ soil) with only two crops. The plant species has the potential of mining 100 $kg\ Ni\ ha^{-1}$ at many sites worldwide. Therefore, phytomining for Ni could be at least as profitable in Ni-contaminated areas as wheat farming. A list of plant species capable of hyperaccumulating certain metal ions is provided in Table 3. The efficiency of phytoextraction is the product of a simple equation, i.e. biomass $\times$ element concentration in biomass.

An important advantage associated with hyperaccumulators is their ability to grow at elevated external metal concentrations. This allows them to remove greater amounts of metal contaminants in a sustainable way (McGrath et al., 2002). In contrast, heavy metal poisoning and growth retardation prevent metal uptake by roots of non-hyperaccumulating plants

species (Nedelkoska and Doran, 2000). Thus, the metal hypertolerance could be a feasible key plant characteristic required for hyperaccumulation of metals. Phytoremediation seems to be most attractive in situations where: (1) drainage water disposal problems relating to a potentially toxic trace element exist, (2) an economically suitable trace element hyperaccumulator can be grown successfully, and (3) other treatment processes of the drainage water are either unavailable or expensive. In addition, this practice may help reduce the adverse ecological effects concerning disposal of contaminated waters.

Phytoremediation of other types of degraded environments, such as widespread salt-affected soils—occupying at least 20% of the world's irrigated land—has also been found to be an efficient, inexpensive, and environmentally acceptable strategy (Qadir et al., 2001b). As an important category of salt-affected soils, sodic soils are characterized by the occurrence of excess Na^+ to levels that can adversely affect crop growth and yield (Sumner, 1993; Qadir and Schubert, 2002). On such soils, phytoremediation works through plant roots to enhance dissolution of slowly soluble native soil calcite ($CaCO_3$) to provide calcium (Ca^{2+}) to replace Na^+ from the cation exchange sites (Qadir et al., 1996; Batra et al., 1997). Several plant species of agricultural significance have been found to be effective in soil amelioration through phytoremediation. Among these species, Kallar grass (*Leptochloa fusca* (L.) Kunth), sesbania (*Sesbania bispinosa* (Jacq.) W. Wight), and Bermuda grass (*Cynodon dactylon* (L.) Pers.) have emerged as potential phytoremediation crops (Qadir and Oster, 2002). Although this vegetative bioremediation strategy is slower in action than the cost-intensive chemical approach, it has shown to be advantageous in several economic, environment, and agronomic aspects: (1) no financial outlay to purchase chemical amendments, (2) financial or other benefits from crops grown during amelioration, (3) promotion of soil aggregate stability and creation of macropores that improve soil hydraulic properties, (4) better plant nutrient availability in soil during and after phytoremedition, (5) more uniform and greater zone of amelioration in terms of soil depth, and (6) carbon sequestration.

5. Use of saline-sodic drainage waters for irrigation

A major problem with irrigated agriculture is its negative environmental impact. Most current waterlogging and salinity/sodicity problems of irrigated lands and impaired water quality are the consequence of inappropriate management of good-quality irrigation waters. Presently, several arid and semiarid regions have the prevalence of saline and/or sodic groundwaters. At the same time, such agricultural regions are left with limited supplies of good-quality irrigation waters, which are insufficient to meet the crop water requirements for the entire irrigation season. Consequent to such changes in available water resources, these areas have an excess of poor-quality waters together with limited supplies of good-quality waters for irrigation. Indeed this is a 'water excess-water shortage' dilemma.

Drainage from irrigated lands is an inevitable phenomenon, which carries a salt load that always is higher, and sometimes substantially higher, than irrigation water. Before the 1970s, the concern on the part of scientists was limited to salinity effects on crop productivity, its control within the root zone by leaching, and drainage water disposal. During the last two decades, concerns have arisen regarding off-site impacts of irrigation and drainage (Van Schilfgaarde, 1994). Without a viable means of use or disposal, the saline-sodic drainage

waters are turning out to be an environmental burden. However, studies conducted on the re-use of such waters for irrigation have shown promise for crop production systems and soil management (Ayers and Westcot, 1985; Minhas, 1996; Rhoades, 1999; Moreno et al., 2001). The key to this approach would be to grow appropriate salt- and Na^+-resistant crops and to maintain soil permeability in order to control soil salinity and sodicity through leaching. The aim behind the successful use of such waters for irrigation would be: (1) to obtain adequate stand and yield of appropriate salt-resistant crops, (2) to control salinity, sodicity, and waterlogging problems in the soil, (3) to maintain soil hydraulic properties, and (4) to protect water quality for long-term sustainable agriculture. Posnikoff and Knapp (1996) have provided a favorable economic assessment of the potential for re-using saline waters for irrigation.

In order to make the drainage water re-use strategy successful and environmentally acceptable, there is a need to take into consideration several management aspects. For instance, an extra quantity of water—from irrigation or resulting from a predictable rainfall—in excess of that needed for evapotranspiration must be applied as a long-term strategy and done in a manner that does not adversely affect the growing crops. This will prevent excessive accumulation of salts in the root zone. The extra quantity of irrigation water, referred to as leaching requirement, must be able to pass through the root zone. Adequate drainage is an essentiality for obtaining a desired leaching requirement to maintain soil salinity at levels suitable for crop growth. It also keeps the water table sufficiently deep to permit adequate root development, prevents the net upward flow of salt-laden groundwater into the root zone, and permits the movement and operations of farm implements in the fields (Shalhevet, 1994; Rhoades, 1999). Artificial drainage systems must be installed if adequate natural drainage is not available.

The re-use of saline and saline-sodic drainage waters, which may be feasible based on research and farmers' experiences, would reduce its volume. Minimizing the volume of irrigation water applied in the first place would also reduce the drainage volume and minimize the leaching fraction. Minimizing the leaching fraction (1) maximizes the precipitation of applied Ca^{2+}, HCO_3^-, and SO_4^{2-} salts as calcite and gypsum in the soil, and (2) minimizes the pickup of weathered and dissolved salts from the soil (Rhoades et al., 1974). These changes in salt loading are predictable based on the inorganic chemistry of mixed salt solutions. For instance, the salt load from the root zone on an annual basis can be reduced from about 2 to 12 $Mg\,ha^{-1}$ by reducing leaching fraction from 0.3 to 0.1, respectively (Oster and Rhoades, 1975). The extent to which leaching and drainage can be minimized is limited by (1) salt resistance of the crops being grown, (2) salinity and sodicity of the irrigation water, (3) irrigation system distribution uniformity, and (4) variability in soil infiltration rates. In most irrigation projects, the drainage volumes can be reduced appreciably without harming crops or soils, especially with improvement in irrigation management (Van Schilfgaarde, 1976; Rhoades, 1999). These drainage volumes may be further reduced by re-using drainage waters again for irrigation.

In addition to agronomic and environment aspects, economic incentives also promote re-use of drainage waters for crop production systems. Such incentives can be designed to motivate near-term reductions in effluent and long-term investments that will reduce the volume of effluent generated per unit of agricultural production (Knapp, 1999). Increasing block-rate prices and salt-load surcharges can motivate farmers to improve water

management practices and reduce unnecessary deep percolation. Such pricing would motivate farmers to choose irrigation methods that will reduce regional salt loads. For instance, Wichelns et al. (1996) reported reductions in water applications ranging from 9% on tomato (*Lycopersicon esculentum* Mill.) fields to 25% on cotton fields as a result of implementing block-rate prices and other economic incentives.

There are several approaches and practices that involve the use of saline and/or sodic waters for irrigation (Rhoades, 1989; Tanji, 1997; Oster, 2000; Shannon and Grieve, 2000). Rhoades (1999) has provided valuable insight into different drainage water re-use strategies where: (1) insufficient supplies of good-quality irrigation waters exist. Generally, two strategies are common to use saline drainage waters to supplement good-quality irrigation waters, e.g. cyclic/serial, and blended use. These drainage water re-use options presuppose the availability of two water sources, i.e. one of good-quality (nonsaline-nonsodic) and the other of poor quality (saline and/or sodic). (2) Only drainage waters are available. Under such conditions, sequential re-use is the strategy.

The cyclic strategy of the drainage water re-use involves irrigation of salt-sensitive crops with good-quality water followed by irrigation of salt-resistant crops with saline water. The good-quality water is usually the developed water supply of the irrigation project or a good amount of predictable rainfall. The poor-quality water is the drainage water generated in the project. Typically, the good-quality water is also used before planting and during critical growth stages of the salt-resistant crops. Saline water is usually used after seedling establishment of the salt-resistant crops. After the harvest of a salt-resistant crop, an irrigation with low-salinity water is applied to the field to leach undesirable salts from the upper portion of the soil profile to provide an environment suitable for the growth of the subsequent salt-sensitive crop. The serial strategy is executed by developing (1) a crop rotation plan that can make best use of the available low- and high-electrolyte waters, and (2) an irrigation plan for the entire crop rotation duration that can be based on crop tolerance against irrigation water salinity and sodicity, and salt sensitivity of the selected crops at different growth stages. Field studies conducted in several areas of the world involving the cyclic re-use of saline drainage waters for irrigation have demonstrated that this strategy is sustainable on a broad range of soils, provided the problems of soil crusting, poor aeration and tilth are dealt with optimum management (Rhoades, 1989; Sharma and Rao, 1998; Oster, 2000).

Blending consists of mixing good- and poor-quality water supplies before or during irrigation. A prerequisite for blending is a controlled way of mixing both the water supplies. According to Shalhevet (1984), there may be two ways of blending, network dilution and soil dilution. With network dilution, water supplies are blended in the irrigation conveyance system, which then essentially needs a facility to be built for blending. In case of soil dilution, the soil acts as the medium for mixing water of different qualities. According to the availability, different water qualities are altered between or within an irrigation event. Although blending saline drainage water with good-quality irrigation water has been practiced in several countries (Ghafoor et al., 1991; Minhas, 1996; Oster et al., 1999), there can be significant and undesirable impacts because of degraded water quality on down-stream users (Rhoades, 1999). This is particularly important when a highly saline water is one of the blending components. In such a case, there may be a potential loss of consumable water. On the other hand, the cyclic strategy provides the means of isolating salinity impacts to a more local, and shorter time interval and takes advantage of an increasing salt resistance of

plants as they mature. Blending results in greater surface soil salinity over time, which affects seedling establishment and crop yield, and so less opportunity to grow high economic value salt-sensitive crops. However, this strategy may be more practical for situations where drainage water or shallow groundwater is not too saline for the crops.

The sequential strategy involves applying the relatively better quality water to the crop with a lower salt resistance, then using the drainage water from that field – obtained through a tile drainage system – to irrigate crops that have the capability to withstand greater concentrations of salts. The simplest management method is to sequentially use drainage water on fields located down-slope from those where the drainage water is collected. There is no fixed number of times for which a drainage water may be sequentially used. Such usage depends on the salinity, sodicity, and concentration of toxic minor elements in the drainage water, volume of water available, and the economic value and acceptable yield of the crop to be grown with the water. The long-term feasibility of drainage water re-use for irrigation would likely be increased if implemented on a regional scale rather than on a farm scale. Grattan and Rhoades (1990) have provided a schematic presentation of regional drainage water re-use strategy. Regional management of drainage water permits its re-use in dedicated areas so as to localize the impacts of its use while other areas, such as up-slope areas, can be irrigated solely with better quality water. The second-generation drainage water from the primary re-use area may be discharged to other dedicated re-use areas where even more salt-resistant crops can be grown successfully. Ideally, regional coordination and cost-sharing among growers should be undertaken in such a re-use system.

The anticipated future scenario of good-quality water shortage suggests that the time has come for the appropriate management of poor-quality water resources, i.e. to intercept, isolate, and re-use drainage waters within the regions where they are generated. The suitability of drainage waters for irrigation depends very much on the relative need and economic and environmental benefits that can be derived compared to other alternatives and on the specific conditions of use. A crucial management decision before implementing the re-use of drainage water is selection of a suitable plant species (Table 4). Production systems based on salt-resistant forage crops and grasses using saline and/or sodic irrigation

Table 4
Important aspects of crop selection for areas under irrigation with saline and/or sodic drainage waters

Crop selection criterion	Crop response
Market demand or utilization at farm	Essential
Resistance to ambient water and soil salinity/sodicity	Essential
Boron and chloride resistance, if needed	Yes
Resistance to heavy metals, if needed	Yes
Resistance to periodic inundation, if needed	Yes
Compatibility with human or animal diet	Essential
Field management-related aspects	Easy sowing, grazing or harvesting, and other cultural operations
Susceptibility to insect pest and diseases	Undesirable
Crop quality under saline/sodic water irrigation	Improve
Requirement for fertilizers, other chemicals	Less
Compatibility with crop rotations of the region	Essential

waters may be sustainable (Kaffka et al., 2002). The objective of the forage production systems would be to provide a year-round supply of high-quality feeds suitable for grazing and economic weight gains in cattle or sheep, or alternatively for sale to dairy farms as ensilage or hay. If a livestock production system is based on the re-use of drainage water, it can transform such drainage waters from an environmental burden into an economic asset. These production systems can reduce the amount of water that must be disposed off and provide the incentive to install needed artificial drainage systems to sustain and improve soil quality. Oster et al. (1999) have demonstrated through preliminary experiments that Bermuda grass could be grown with saline-sodic waters having EC around 17 dS m^{-1} and SAR exceeding 25. These authors also provided information regarding growth habit, salt resistance characteristics, average root zone salinity at 70% yield, and leaching requirement of several forages when irrigated with a water having salinity of 10 dS m^{-1} and SAR of 15.

6. Conclusions

Being so fundamental to the social, economic, and environmental sustainability of different regions, water is a strategic resource particularly for water-starved countries where more than 40% of the present global population lives. Owing to the use of different criteria, there has been a skepticism concerning potential availability of water for food production to meet the future needs of the expanding population. However, there should be no anxiety in any region of the world, including Middle East and Africa, about the availability of water for drinking and domestic use and for almost all industrial and service sector uses. These sectors hardly use 25% of the freshwater resources. Agriculture is the largest single user of water with about 75% of freshwater being used for irrigation. Whenever the demand for freshwater increases a competition among municipal, industrial, and agricultural sectors often ends up in a decreased allocation to agriculture. This phenomenon is expected to continue leaving less and less freshwater for agricultural use, rather it is expected to intensify in less developed, arid region countries that already suffer from water, food, and health problems. This scenario reveals that agricultural water management must be coordinated with, and integrated into, the overall water management of the water-starved countries. We believe that the following strategies may serve as the essential components of sustainable agricultural water management in such countries: (1) potential use of ‘virtual water’ as a global solution to regional deficits, (2) improvement in current efficiencies of agricultural water use and conservation, (3) development of economically and environmentally acceptable methods for protection and improvement of water and soil quality, and (4) re-use of saline and/or sodic waters in certain dedicated areas. Since there is no single way to cope with the freshwater shortage, suitable combinations of these strategies will help resolve the water crisis, or at least will decrease its intensity in water-starved countries.

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